

1. CONTEXT

We are building a **SYNAPSE Medical Education Platform** (part of Aethex). We need to add a new section called **"AI Presentation Generator"** where users type any medical topic and the AI generates a **full, professional, exam-ready presentation** — rendered directly on the website.

■ **KEY NOTE:** After reading one SYNAPSE presentation, students should never need to look elsewhere for exam prep on that topic.

2. THE PROBLEM WITH CURRENT OUTPUT

The current presentation output is too basic — plain text slides with bullet points. Users need:

- Professional, visually rich slide design
- Auto-generated diagrams (anatomy, flowcharts, mind maps)
- Exam-focused FAQ and quick reference sections
- Complete coverage so students have zero reason to open another resource

3. REQUIRED SLIDE STRUCTURE (12 Slides)

Every generated presentation must contain the following slides:

#	Slide Title	Key Requirements
1	Title Slide	Topic name large & bold, SYNAPSE subtitle, slide count, dark navy/teal gradient hero
2	Overview & Clinical Roles	5-6 clinical relevance points, visual concept map (boxes + arrows), Key Insight callout
3	Anatomy / Core Structure	Detailed content + auto-generated labeled SVG anatomy diagram, Key Insight
4	Physiology / Mechanism	Step-by-step how it works + flowchart diagram showing pathway, Key Insight
5	Innervation / Pathways / Classification	Tables or structured comparison + visual pathway map (SVG labeled nodes), Key Insight
6	Clinical Examination / Bedside	Step-by-step exam technique, checklist layout with icons, Key Insight
7	Common Conditions & Differentials	Condition cards in grid, differential diagnosis table, Key Insight
8	Red Flags & When to Refer	Bold warning-style cards with urgency icons, Key Insight

9	Mind Map (Full Page)	Central topic → branches: Anatomy, Function, Innervation, Clinical, Disorders, Red Flags. Rendered as interactive/SVG mind map
10	FAQ — Exam-Style Q&A;	8-10 high-yield exam questions with collapsible answers in flashcard layout
11	Quick Reference / Glossary	8-10 key terms + definitions in two-column grid with numbered badges
12	Summary & High-Yield Takeaways	6-point summary, final Key Insight, SYNAPSE branding footer

4. DIAGRAM REQUIREMENTS (CRITICAL)

Every slide with anatomical, pathway, or process content **MUST have a diagram**.

- **Library:** Use SVG, D3.js, Mermaid.js, or GoJS for rendering
- **Anatomy slides:** Auto-generate labeled SVG anatomical diagrams
- **Process slides:** Flowchart showing step-by-step mechanism
- **Pathway slides:** Node-graph with labeled connection arrows
- **Mind Map:** Real visual node-graph (NOT bullet points)
- **Concept Maps:** Per-slide, small visual in top-right corner — boxes with arrows
- Diagrams must be **medically accurate in layout, labeled, and clean**

■ **KEY NOTE:** A user searching "Liver" must get a real liver anatomy SVG — not a stock image or placeholder.

5. DESIGN & UI REQUIREMENTS

Element	Specification
Background	#0A0F2C (dark navy)
Accent Color	#00BCD4 (teal)
Text Color	White on dark slides
Fonts	Inter / DM Sans (body), Space Grotesk (headings)
Header Bar	"MEDICAL EDUCATION SYNAPSE" on every slide
Slide Counter	Badge bottom-right (e.g. 3 / 12)
Key Insight Box	Teal left border, lighter background, italic text
Navigation	Left/right arrows + dot indicators
Responsive	Works on both mobile and desktop
PDF Export	Optional: html2canvas + jsPDF export button

6. TECH STACK

- **Frontend:** React or Next.js (match current site stack)
- **AI Engine:** Anthropic Claude API — model: claude-sonnet-4-20250514
- **Diagrams:** Mermaid.js for flowcharts/mind maps + custom SVG for anatomy
- **Slide Renderer:** Custom React component (NOT a third-party library like reveal.js)
- **PDF Export (bonus):** html2canvas + jsPDF

7. USER FLOW

Step 1 — User lands on the "AI Presentation" section

Step 2 — Sees search bar: *"Enter any medical topic..."*

Step 3 — Clicks "**Generate Presentation**"

Step 4 — Loading state: animated skeleton slides with "SYNAPSE is preparing your session..."

Step 5 — Slides render fully with all diagrams and sections

Step 6 — User navigates with arrows or dot indicators

Step 7 — Optional: Download as PDF button

8. CLAUDE API PROMPT STRUCTURE

Use this system prompt when calling the Anthropic API:

```
You are SYNAPSE, a medical education AI. Generate a complete, exam-ready medical education presentation on the topic: [USER TOPIC]. Return a JSON object with an array called "slides". Each slide must have: - slideNumber (int) - title (string) - keyInsight (string) - bullets (array of strings, 5-6 points) - conceptMapNodes (array of {id, label, connections[]}) - diagramType ("anatomy"|"flowchart"|"pathway"|"mindmap"|"table"|"none") - diagramData (structured data for rendering the diagram) - faq (array of {question, answer} - FAQ slide only) - glossary (array of {term, definition} - glossary slide only) - conditions (array of {name, features, clinicalClue} - conditions slide only) Make content high-yield, exam-focused, and clinically relevant. Never return plain paragraph text - always return structured JSON data.
```

9. FINAL GOAL

■ **KEY NOTE:** A user searches "Liver" and gets a full 12-slide professional presentation with real diagrams, mind map, FAQ, and glossary — so complete that they never need a textbook for that topic. This is the core value of SYNAPSE.

The SYNAPSE AI Presentation Generator is the flagship feature of the Aethex medical education platform. It must be fast, beautiful, and medically accurate. Every presentation generated must feel like it was designed by a medical educator — not produced by a chatbot.